

Incorporating a Chronology Response into the Prediction of Leaf Appearance Rate in Winter Wheat

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The prediction of leaf appearance rate (LAR) is an important part of many crop simulation models. Most wheat simulations models assume that LAR is affected by temperature and photoperiod. This assumption ignores the fact that seed reserves contribute to a greater LAR of the first two leaves and that the LAR of subsequent leaves decreases as a result of an increase in the distance that each leaf primordium must extend before it appears. The objective of this study was to develop a generalized LAR chronology response function [$f(C)$] for wheat that takes into account seed reserves and the increasing distance from the meristem to the whorl for later appearing leaves. This chronology response function was then incorporated into an existing LAR model [Wang and Engel (WE) model; Wang and Engel, 1998, *Agricultural Systems* 58: 1–24]. This function varied from 0 to 1, being equal to 1 for the first two leaves due to seed reserves, and decreasing (taking the form of a power law) for subsequent leaves. Data from a growth chamber (two cultivars) and several field experiments (four cultivars, two years and eight sowing dates) at Lincoln, Nebraska, USA, were used as independent data to test three LAR models (Miglietta model, Miglietta, 1991, *Climate Research* 1: 145–150; WE model; and modified WE model). Predictions of the main stem Haun stage, both in the growth chamber and in the field, were greatly improved by incorporating $f(C)$ into the Wang and Engel model. The root mean square error for the field data was 1.1, 0.7, and 0.3 leaves for the Miglietta model, the Wang and Engel model, and the modified Wang and Engel model, respectively.

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Key words: Leaf appearance, seed reserves, model, temperature, photoperiod, *Triticum aestivum*.

INTRODUCTION

The calculation of leaf appearance rate (LAR) is an important component of many crop simulation models. The integration of LAR over time gives the number of emerged leaves on the plant main stem; this is an excellent measure of plant development. The number of emerged leaves in wheat (*Triticum aestivum* L.) is often represented by the Haun stage (HS), which is the number of fully expanded leaves plus a ratio of the length of the expanding leaf to the penultimate leaf (Haun, 1973). In wheat, the HS has been related to tiller appearance (Klepper *et al.*, 1982; McMaster *et al.*, 1991; Rickman and Klepper, 1991), and to the timing of certain key developmental stages, such as double ridge, terminal spikelet and anthesis (Brooking *et al.*, 1995; Calderini *et al.*, 1996; Robertson *et al.*, 1996; Jamieson *et al.*, 1998). Accurately modelling the appearance of individual leaves and the rate of leaf area expansion also has an impact on modelling of light absorption by the canopy, canopy photosynthesis and, therefore, accumulation of dry

matter and yield (Amir and Sinclair, 1991; Hodges and Ritchie, 1991; McMaster *et al.*, 1991).

Temperature and photoperiod are the two major factors that affect LAR in wheat (Cao and Moss, 1989a, b; Kirby, 1995; Slafer and Rawson, 1995a, 1997). Minor factors, such as soil compaction, depth of sowing, vernalization, solar radiation, water and nutrient availability, salinity and CO₂ have also been reported to affect LAR (Kirby, 1995; Wilhelm and McMaster, 1995; Rawson *et al.*, 1998), but these factors generally have little or no impact on values of LAR measured in the field (Miglietta, 1991b; Kirby, 1995). Consequently, most wheat simulation models have only a temperature and a photoperiod response function in the equation that calculates LAR (e.g. Volk and Bugbee, 1991; Wang and Engel, 1998).

Results from controlled-environment and field experiments consistently show that there is a time or chronology factor that affects LAR in wheat besides temperature and photoperiod, with LAR decreasing for later appearing leaves. Data from Slafer and Rawson (1997) showed a positive curvilinear relationship between number of emerged leaves and days after emergence for three wheat cultivars ('Condor', 'Rosella' and 'Capelle Desprez') grown at constant temperature and photoperiod. In field experi-

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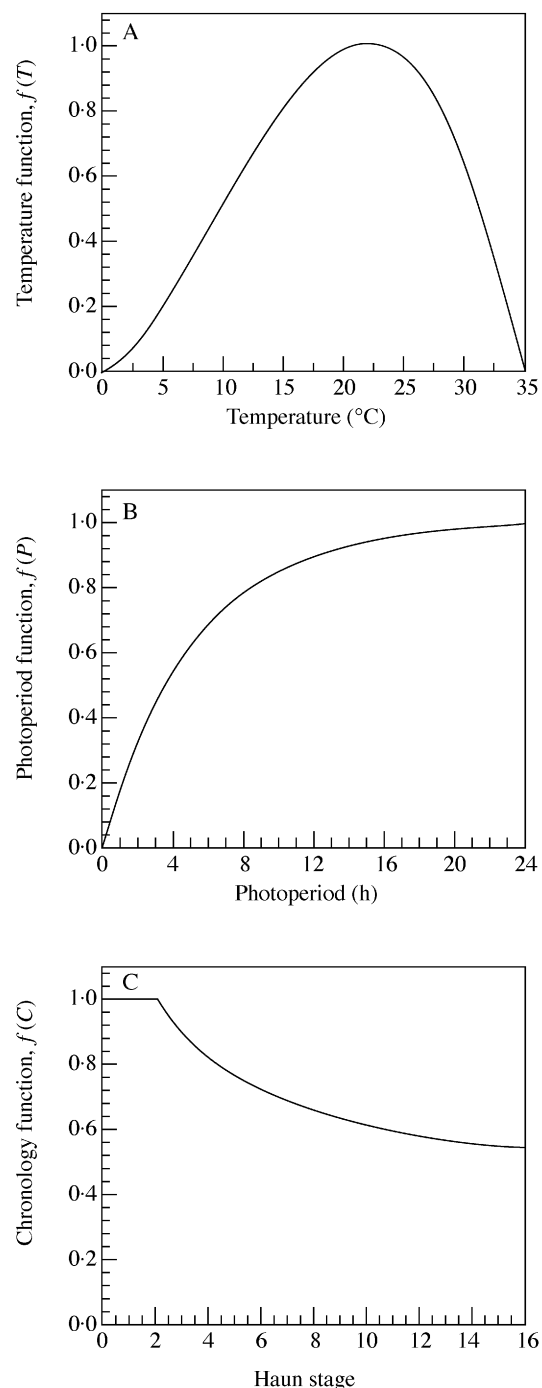


FIG. 1. The wheat leaf appearance rate response functions to temperature [$f(T)$, eqn (2)] (A), photoperiod [$f(P)$, eqn (4)] (B) and chronology [$f(C)$, eqn (5)] (C).

ments with different cultivars, this relationship between number of emerged leaves and thermal time, TT, (Kirby *et al.*, 1985; Calderini *et al.*, 1996; Miralles *et al.*, 2001; González *et al.*, 2002) or photothermal time (Kirby, 1995) was also found to be curvilinear. A bilinear relationship between HS and TT, with HS in the range of four to eight leaves as the breaking point, was observed in other field

studies (Baker *et al.*, 1986; Boone *et al.*, 1990; Hotsonyame and Hunt, 1997), with the slope between HS and TT being lower for leaves appearing after eight emerged leaves. There are two reasons for a decrease in LAR throughout the growing season: (1) seed reserves are responsible for a greater LAR of the first two leaves (Peterson *et al.*, 1989); and (2) as the number of emerged leaves increases, leaves take more time to appear because the distance that each primordium has to extend before appearing at the whorl increases for each subsequent leaf (Gallagher, 1979; Skinner and Nelson, 1995).

The rate of change of leaf appearance as a function of leaf number has been considered by several authors. Miglietta (1991a) assumed an exponential decrease of LAR with number of emerged leaves. This effect was also considered by Jamieson *et al.* (1995) who assumed that the phyllochron (TT for successive leaves to appear) increases throughout the growing season (75 °Cd for the first two leaves, 100 °Cd for leaves 2–8, and 130 °Cd for leaves greater than 8). Neither of these models, however, considers the effect of seed reserves on LAR. Xue (2000) compared several LAR models to predict HS of four winter wheat cultivars at Lincoln, Nebraska, USA, and found that the model of Wang and Engel performed better than the models of Miglietta (1991a) and Jamieson *et al.* (1995). One reason for the poorer performance of the Miglietta and Jamieson models in Xue's study may be that the non-linear assumptions in these models are poor descriptors compared with the non-linear approach used in the Wang and Engel (WE) model. The WE model, which does not have a chronology response function, had an error of about one leaf, but produced systematic errors (sometimes greater than one leaf), with underpredictions of HS at early stages (HS = two to four leaves) and overpredictions at late stages (HS = 10+ leaves) (Xue, 2000). The systematic errors with the WE model suggest that there is a potential for improvement of this model.

Most existing LAR models do not deal with the rate of change of leaf appearance with leaf number, nor do they consider the effect of seed reserves on LAR. These deficiencies provide a rationale for improving the predictions of LAR in wheat. The objective of this study was to develop a generalized LAR chronology response function for winter wheat that takes into account seed reserves and the rate of change of leaf appearance with leaf number, and incorporate this function into an existing LAR model (WE model; Wang and Engel, 1998). It was hypothesized that the inclusion of an LAR chronology response function and the effect of seed reserves would improve the predictions of the LAR in winter wheat.

MATERIALS AND METHODS

The Wang and Engel (WE) LAR model

The general form of the WE model to calculate LAR is:

$$R_{LA} = R_{LA,max} f(T) f(P) \quad (1)$$

where R_{LA} is the daily leaf appearance rate (leaves d^{-1}), $R_{LA,max}$ is the maximum daily leaf appearance rate (leaves d^{-1}) under optimum temperature and photoperiod, and $f(T)$

and $f(P)$ are dimensionless temperature and photoperiod response functions (0–1) for LAR.

The value of $R_{LA,max}$ [maximum daily leaf appearance rate (leaves d^{-1}) under optimum temperature and photoperiod] for the four winter wheat cultivars used in this study ('Arapahoe', 'Karl 92', 'TAM 107' and 'NE 92458') was 0.24 leaves d^{-1} (Xue, 2000). These cultivars were chosen because they have different pedigrees, different developmental patterns, and were used in the study by Xue (2000). Arapahoe and Karl 92 are widely grown in Nebraska and Kansas, respectively. When planted in Nebraska, Karl 92 is one of the earliest flowering cultivars and Arapahoe is medium to medium-late in flowering. TAM 107 was developed in Texas and is widely grown in the southern Great Plains. Like Karl 92, it is a very early flowering cultivar in Nebraska. NE92458 was an elite experimental line developed in Nebraska, and is medium in its flowering date. The value of $R_{LA,max}$ for the four winter wheat cultivars used in this study was based on the results of Xue (2000). Different cultivars do not necessarily have different values of $R_{LA,max}$. For instance, Jame *et al.* (1999) and Yan and Hunt (1999) derived values of $R_{LA,max}$ for several wheat cultivars used in the work of Cao and Moss (1989a, b), and found that the winter wheat cultivars 'Yamhill', 'Stephens' and 'Tres' had the same $R_{LA,max}$. Differences among cultivars are probably more evident in other developmental parameters, such as main stem final leaf number, than $R_{LA,max}$.

The temperature function $f(T)$ (Fig. 1A) is:

$$f(T) = [2(T - T_{min})^\alpha (T_{opt} - T_{min})^\alpha - (T - T_{min})^{2\alpha}] / (T_{opt} - T_{min})^{2\alpha} \text{ for } T_{min} \leq T \leq T_{max}$$

$$f(T) = 0 \text{ for } T < T_{min} \text{ or } T > T_{max} \quad (2)$$

$$\alpha = \ln 2 / \ln [(T_{max} - T_{min}) / (T_{opt} - T_{min})] \quad (3)$$

where T_{min} , T_{opt} and T_{max} are the cardinal temperatures (minimum, optimum and maximum) for LAR, and T is the mean daily temperature calculated from the 24 h temperature (Xue, 2000). The same cardinal temperatures used by Xue (2000) were used in this study, i.e. 0, 22 and 35 °C, respectively.

The photoperiod function $f(P)$ (Fig. 1B) is:

$$f(P) = 1 - \exp[-\omega(P - P_c)] \quad (4)$$

where P is the actual photoperiod (h), P_c is the critical photoperiod (h) below which LAR is zero, and ω is a photoperiod sensitivity coefficient (h^{-1}). The photoperiod, including civil twilight (when the sun is from 0 to 6° below the horizon), was calculated using the algorithm of Kiesling (1982). The coefficient P_c in eqn (4) was assumed to be 0 (Volk and Bugbee, 1991; Xue, 2000), and ω was 0.2 h^{-1} for all cultivars used in this study (Xue, 2000).

The number of emerged leaves on the main stem, expressed as the Haun stage (HS), was calculated by accumulating daily LAR values (i.e. at a daily time step) starting at emergence, i.e. $HS = R_{LA,max}$.

The LAR chronology response function

The LAR chronology response function [$f(C)$] was defined as the rate of change of LAR as a function of time. Time, in this context, was measured as the number of emerged leaves, represented by the HS.

The first two leaves in wheat have the highest LAR owing to seed reserves (Peterson *et al.*, 1989), which represent the source of carbohydrates and nutrients before emergence and during the first days after emergence (Williams, 1960). Therefore, it was assumed that $f(C) = 1$ for the first two leaves.

As the number of emerged leaves increases, the distance that each leaf primordium must extend to appear at the whorl increases for each subsequent leaf (Gallagher, 1979; Skinner and Nelson, 1995); this in turn decreases LAR. The increased distance for each subsequent leaf to appear, combined with a rapid exhaustion of seed reserves when the first two leaves have expanded (Peterson *et al.*, 1989), led to the assumption that the decrease in LAR after Haun stage 2 followed a power law. The LAR chronology response function was then defined as:

$$f(C) = 1 \quad \text{if } HS < 2$$

$$f(C) = (H/2)^b \quad \text{if } HS \geq 2 \quad (5)$$

where H is the Haun stage and b is a sensitivity coefficient (unitless) representing the decrease in $f(C)$ for each unit of increase in HS (Fig. 1C). The coefficient b was set to -0.3 and was assumed to be the same for all cultivars, based on a power function used by Yin and Kropff (1996) to fit HS vs. time (days) at constant temperature in rice (*Oryza sativa* L.), which is similar to wheat in morphology and growth habit. Yin and Kropff (1996) fitted a power function to the relationship between Haun stage (H) and days after emergence, D , ($H = D^a$). The first derivative of this function, i.e. $dH/dD = aD^{a-1} = aD^b$, is a chronology function. The value of coefficient, a (0.7), was calculated as the mean value presented in Yin and Kropff (1996) for several rice cultivars. The value of coefficient b ($a - 1$) is then -0.3, and represents the shape coefficient of the chronology function.

The modified WE LAR model

The WE model [eqn (1)] was modified by including the LAR chronology function eqn (5):

$$R_{LA} = R_{LA,max} f(T) f(P) f(C) \quad (6)$$

where $R_{LA,max}$ is the maximum daily leaf appearance rate (leaves d^{-1}) of the first two leaves under optimum temperature and photoperiod; all other terms have been defined previously. Note that in eqn (6), when $f(T)$ and $f(P)$ are unity, $R_{LA} = R_{LA,max}$ only for the first two leaves, because when $HS > 2$, $f(C) < 1$. Therefore, the interpretation of $R_{LA,max}$ in eqn (6) differs to that in the original WE model [eqn (1)], where $R_{LA} = R_{LA,max}$ any time $f(T)$ and $f(P)$ are unity. However, the biological meaning of $R_{LA,max}$ in eqn (6) is maintained.

In eqn (6), $f(T)$, $f(P)$ and $f(C)$ were combined in a multiplicative model as in the original WE model [eqn (1)]. There are different approaches to combining the different

factors that affect plant development: limiting factor models (e.g. Ritchie, 1991); additive models (e.g. Perry *et al.*, 1987; Summerfield *et al.*, 1991; Saarikko and Carter, 1996; Jamieson *et al.*, 1998); multiplicative models (e.g. Angus *et al.*, 1981; Weir *et al.*, 1984; Ewert *et al.*, 1996; Cao and Moss, 1997); and a combination of both additive and multiplicative approaches (e.g. Yan and Wallace, 1998). The multiplicative approach appears more realistic from a biological point of view because interactions among factors that control development have been verified both in the field and in controlled environment experiments with wheat (Slafer and Rawson, 1994, 1995b; Ortiz-Ferrera *et al.*, 1995; González *et al.*, 2002).

The value of $R_{LA,max}$ in eqn (6) was estimated from the value for $R_{LAR,max}$ used in eqn (1) that was derived by Xue (2000). The value of $R_{LA,max}$ derived by Xue (2000) did not include the effect of seed reserves, and it represents the $R_{LAR,max}$ of a plant with an average of approx. seven leaves because data for HS > 2 and up to 12 leaves were used in its derivation. Peterson *et al.* (1989) found that LAR of the first two leaves is, on average, 32 % greater than the mean LAR of subsequent leaves. Therefore, $R_{LA,max}$ in eqn (6) was estimated as $R_{LA,max} = (1.32)(0.24 \text{ leaves d}^{-1}) = 0.32 \text{ leaves d}^{-1}$.

The Miglietta LAR model

The LAR algorithm used by Miglietta (1991a) was compared with the original and modified WE models because it also has a chronology function. The number of emerged main stem leaves (H) in the Miglietta (1991a) model is calculated by:

$$H = \{1 - \exp[-0.03(P-4)]\}/0.03 \quad (7)$$

where P is the daily integration of P_r (the daily rate of leaf primordia initiation), i.e.

$$P = \sum P_r \quad (8)$$

where

$$P_r = -0.038 + 0.0149T \text{ for } T > 2.55 \text{ }^{\circ}\text{C} \\ P_r = 0 \text{ for } T \leq 2.55 \text{ }^{\circ}\text{C} \quad (9)$$

The model by Jamieson *et al.* (1995) also takes into account chronology factors through an increase in phyllochron with leaf number, but it was not used in this study because it produces similar results to the Miglietta model for the same cultivars in the same location (reported by Xue, 2000).

Growth chamber experiment

A growth chamber experiment was conducted to measure the number of emerged leaves of winter wheat plants under constant temperature and photoperiod. When temperature and photoperiod are held constant, and with the assumption that only these two factors affect LAR, HS should increase linearly as a function of time (calendar days). If there is a departure from this linearity, then chronological factors (seed reserves and the increasing

distance that each leaf primordium must extend to appear) influence LAR.

The growth chamber was located at the University of Nebraska, Lincoln, USA, and the experiment was conducted from 2 Nov. 1999 to 30 Apr. 2000. Unvernalized plants of two winter wheat cultivars (Arapahoe and Karl 92) were grown at $25 \pm 0.5 \text{ }^{\circ}\text{C}$ and 20 h photoperiod throughout the entire growing period. Unvernalized plants were chosen so that a large number of leaves formed on the main stem. The final leaf number (FLN) at the end of the experiment was 19 leaves for Karl 92 and 21 leaves for Arapahoe. Eight pots (17 cm in diameter $\times$ 40 cm in height), with two plants per pot, were used for each cultivar (which were the treatments). Plants were grown with adequate water and nutrients. A completely randomized experimental design with eight replications was used; each pot (two plants) was an experimental unit. Main stem HS on all plants was measured daily until plants had about ten fully expanded leaves, and twice or three times a week thereafter.

Air temperature and photosynthetic photon flux density (PPFD) inside the growth chamber were measured throughout the growing period using five type K thermocouples and a quantum sensor (model LI-190SA; Li-Cor, Inc., Lincoln, NE, USA) respectively, connected to a data logger (model CR10-X; Campbell Scientific, Inc., Logan, UT, USA). Three thermocouples were located at the pot level and two thermocouples were 80 cm above the pots. The quantum sensor was placed in the central part of the growth chambers 60 cm above the pot level (1 m from the ceiling). Temperature and PPFD were sampled every 30 s, and the mean value and s.d. were recorded every 15 min. Temperature was homogeneous horizontally and vertically, and PPFD was about $650 \mu\text{mol m}^{-2} \text{ s}^{-1}$ throughout the experiment.

Field experiments

A series of field experiments was carried out to evaluate the influence of varying temperature and photoperiod on the number of emerged leaves. These experiments were conducted at the Havelock Farm, Department of Agronomy and Horticulture, University of Nebraska—Lincoln, Lincoln, NE, USA ($40^{\circ}51'N$, $96^{\circ}36'W$, 347 m a.s.l.) during two growing seasons, 1999–2000 and 2000–2001. In the 1999–2000 growing season, two winter wheat cultivars (Arapahoe and Karl 92) and six sowing dates (13 and 29 Sep. 1999, 13 Oct. 1999, 15 Nov. 1999, 29 Mar. 2000 and 2 May 2000) were used. In the 2000–2001 growing season, four winter wheat cultivars (Arapahoe, Karl 92, TAM 107 and NE 92458) and two sowing dates (29 Sep. 2000 and 12 Oct. 2000) were used.

The experiments were in a randomized complete block design with four replications. Each plot was $1.2 \times 2.4 \text{ m}$, with four rows in each plot in an east–west direction. The row spacing was 0.30 m and the plant density was about 200 plants m^{-2} . Plants were grown under rain-fed conditions and plots were well fertilized before sowing. At sowing, soil tests in the 0–60 cm layer indicated 18.1 ppm NO_3 , 23.1 ppm P and 365.0 ppm K in the 1999–2000 growing season, and 25.3 ppm NO_3 , 21.9 ppm P and 256.5 ppm K in the

TABLE 1. Monthly mean air temperature and total monthly precipitation during two growing seasons at Lincoln, NE, USA

Season	Month	Air temperature (°C)	Precipitation (mm)
1999–2000	October	12.0	1
	November	8.4	24
	December	0.2	15
	January	–1.9	2
	February	3.0	40
	March	7.2	20
	April	11.4	38
	May	19.4	46
	June	22.1	104
Mean/total		9.1	290
2000–2001	October	13.8	70
	November	0.1	27
	December	–8.9	16
	January	–2.7	23
	February	–5.5	46
	March	2.2	34
	April	13.2	68
	May	17.7	203
	June	22.0	71
Mean/total		5.7	558

2000–2001 growing season. The soil at the experimental site is a Butler silt loam (fine, montmorillonitic, mesic Abruptic Agriaquoll).

Emergence was measured in four 0.5-m row lengths in each plot by counting the number of emerged plants on a daily basis. Emergence date was defined as the day when 50 % of the plants were emerged. Two days after emergence, six plants located in the two centre rows (three plants per row) in each replication were randomly selected and tagged using coloured wires. These plants were used to measure the main stem HS throughout the entire experiment. The frequency of HS measurements varied from daily to 10-d intervals, depending on sowing date, year and weather conditions. FLN at anthesis was also recorded. Hourly meteorological data (air temperature, precipitation and solar radiation) were measured at an automated weather station 300 m from the field site.

Model evaluation

The LAR chronology response function $f(C)$ [eqn (5)] was evaluated using data from the growth chamber and field experiments for cultivars Arapahoe and Karl 92. Growth chamber HS data were collected daily. Field HS data were collected on 6 [day of year (DOY) = 97] and 18 Apr. 2000 (DOY = 109) from plants sown on 13 Sep. 1999, 29 Sep. 1999, 13 Oct. 1999, 15 Nov. 1999 and 29 Mar. 2000. During the period between these field observations, daily mean temperature varied from 2.8 to 16.9 °C (mean = 9.2 °C), and the HS increased by about one leaf. The LAR (leaves d^{–1}) was calculated as the difference in HS between the two observations divided by the difference in time (d). LAR data were normalized by dividing each LAR value by the LAR

calculated for the two lowest HS, which was between 1.0 and 2.0 leaves.

The values of HS predicted by the WE model [eqn (1)], the modified WE model [eqn (6)] and the Miglietta model [eqn (7)] were compared with observed values from the growth chamber and field experiments, which were independent data sets. The statistic used to evaluate model performance was the root mean square error (RMSE), calculated as (Janssen and Heuberger, 1995):

$$E = [\sum (p_i - o_i)^2 / n]^{0.5} \quad (10)$$

where E is RMSE, p is the predicted value, o is the observed value and n is the number of observations. Models with the lowest RMSE are considered best.

RESULTS

There was a large variation in environmental conditions during the eight sowing dates in the two field growing seasons. The 1999–2000 growing season was warmer and drier than the 2000–2001 growing season (Table 1). Plants sown in autumn 1999 and autumn 2000 were exposed to daily temperatures below 0 °C during the following winter months (December–February) and to mild temperatures (10–20 °C) in March and April, flowering by early May. Plants sown on 29 Mar. 2000 and 02 May 2000 were exposed to mild temperatures (10–20 °C) at the beginning of the growing period, and to high temperatures (several days with maximum temperatures greater than 35 °C) during May and June, dying at the end of June without flowering. Plants from all sowing dates in the 1999–2000 growing season showed mild to severe symptoms of water stress (leaf rolling during the afternoon) on several days in April and May 2000, whereas no water stress symptoms were observed on any day during the 2000–2001 growing season.

Different environmental conditions in the field and growth chamber resulted in plants whose final leaf number ranged from eight to 20 leaves. A wide range of conditions is represented by these experiments, from unvernallized plants (growth chamber experiment and plants sown in the field on 2 May 2000) to plants that were fully vernalized at different stages (several sowing dates in the field), and from unstressed plants to plants that were water stressed at different developmental stages. This wide range of environmental conditions provides a rich data set to evaluate the three LAR models.

The LAR chronology response function $f(C)$ [eqn (5)] and the observed LAR data, as a function of HS, from the growth chamber and field experiments are illustrated in Fig. 2. There is considerable variability in the growth chamber data (Fig. 2A) probably as a result of the natural variability of the daily HS data. However, the growth chamber data for both cultivars (Arapahoe and Karl 92) show a general trend of decreasing LAR as the number of main stem leaves (HS) increases, and most of the data are spread around the predicted curve. The field observations were closer to the predicted data (Fig. 2B). These results (data from Fig. 2A and B) indicate that the assumptions used to develop $f(C)$ are reasonable.

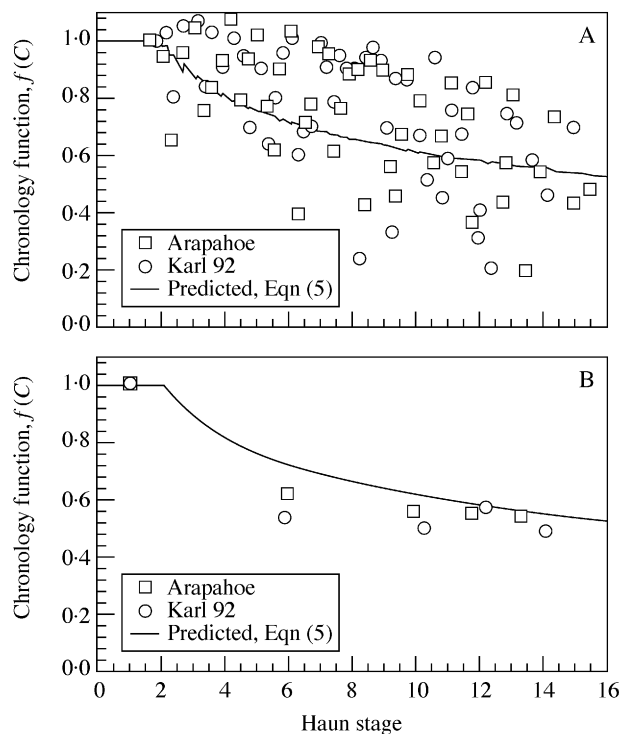


FIG. 2. Observed and predicted leaf appearance rate chronology response functions, $f(C)$ vs. Haun stage for two winter wheat cultivars ('Arapahoe' and 'Karl 92') grown in a growth chamber at a constant temperature (25 °C) and photoperiod (20 h) (A) and in the field during the 1999–2000 growing season (B).

The main stem HS observed and predicted by the three LAR models for Arapahoe and Karl 92 grown in a growth chamber is illustrated in Fig. 3. The observed data show a curvilinear relationship that departs considerably from the linear predictions of the WE model (RMSE = 1.1 leaves for both cultivars). The curvilinear response is captured well by the modified WE model [eqn (6)], which reduced the RMSE to 0.6 and 0.07 leaves for Arapahoe and Karl 92, respectively. The Miglietta model also captured part of the curvilinear response but consistently overpredicted HS for both cultivars when plants had more than about five leaves (RMSE = 0.8 leaves for Arapahoe and 1.7 leaves for Karl 92).

The performance of the three LAR models in simulating the main stem HS for the two winter wheat cultivars on two sowing dates during the 1999–2000 growing season (15 Nov. 1999 and 2 May 2000) and for the four winter wheat cultivars at one sowing date during the 2000–2001 growing season (29 Sep. 2000) are presented in Figs 4 and 5, respectively. Plants on the two sowing dates in the 1999–2000 growing season grew under extreme temperature conditions. For plants sown on 15 November, vernalization took place before emergence (during November and December 1999) and plants flowered in May 2000. Plants sown on 2 May developed during May and June 2000, were not vernalized, and did not flower. The 29 Sep. 2000 sowing date (Fig. 4) represents a normal sowing date for local conditions, with plants producing approx. five leaves before

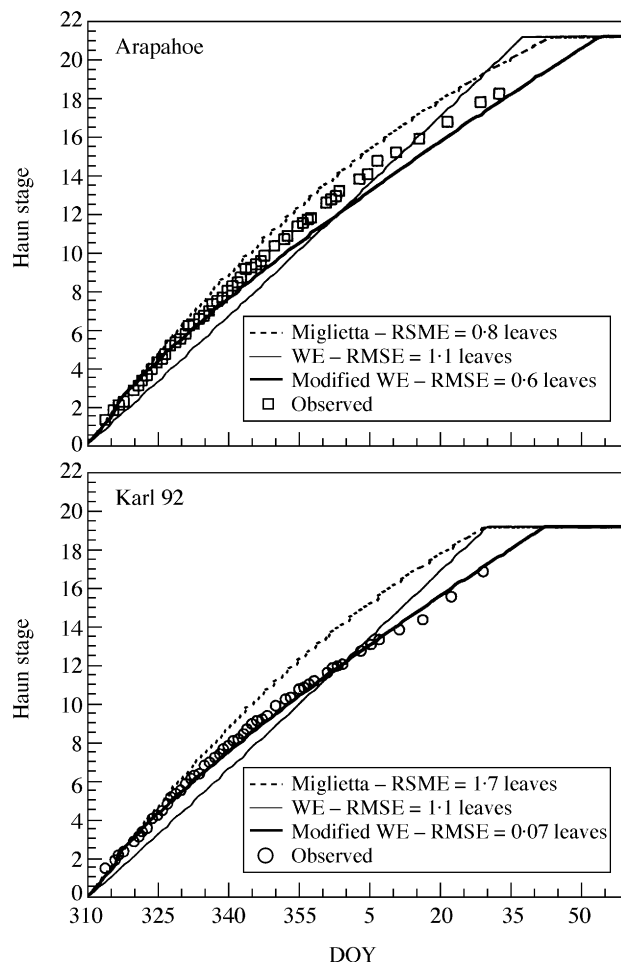


FIG. 3. Observed main stem Haun stage and values predicted by the three LAR models (Miglietta; WE; modified WE) for two winter wheat cultivars ('Arapahoe' and 'Karl 92') grown in a growth chamber at a constant temperature (25 °C) and photoperiod (20 h).

being covered by snow from the end of November to the end of February, and restarting growth in March. The modified WE model was always superior to the other two models in both growing seasons, with a RMSE that ranged from 0.1 to 0.5 leaves. The Miglietta model did not simulate LAR as well as the other models, with a RMSE that ranged from 0.4 to 1.3 leaves. The only exception was for the sowing on 02 May 2000 when it gave better predictions (RMSE = 0.4 for Arapahoe and 0.5 for Karl 92) than the original version of the WE model (RMSE = 0.7 for Arapahoe and 0.9 for Karl 92), but the modified WE model was still better for this sowing date (RMSE = 0.2 for both cultivars). The modified WE model was also superior to the other two models for the other four planting dates in the 1999–2000 growing season and in the second planting date in 2000–2001 (data not shown).

Predicted vs. observed values for main stem HS for all field data (pooling data for different cultivars, sowing dates and years) are presented in Fig. 6. The statistics show a considerable improvement in the predictions of HS with the modified WE model (RMSE = 0.3 leaves). The RMSE was

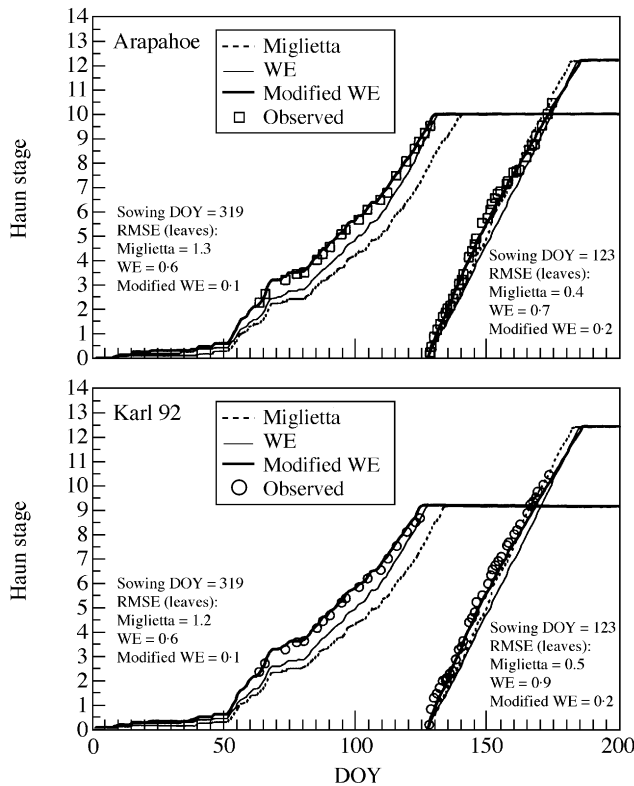


FIG. 4. Observed main stem Haun stage and values predicted by the three LAR models (Miglietta; WE; modified WE) for two winter wheat cultivars ('Arapahoe' and 'Karl 92') sown on 15 Nov. 1999 (DOY = 319) and 2 May 2000 (DOY = 123) in the field at Lincoln, NE, USA.

1.1 leaves for the Miglietta model, and 0.7 leaves for the WE model. The WE model had systematic prediction errors, with underpredictions at low HS ($HS < 8$), good predictions at intermediate HS ($8 < HS < 11$), and overpredictions at high HS ($HS > 11$) (Fig. 5). These systematic errors with the WE model are a result of an $R_{LA,max}$ that represents an average over the entire growing season and does not include the effects of seed reserves and leaf number on LAR. The Miglietta model also had systematic errors, most of them owing to underprediction of HS. To quantify the systematic and unsystematic errors of the model predictions, the mean square error (MSE), i.e. $(RMSE)^2$ was calculated for each model and decomposed into systematic and unsystematic (random) components (Willmott, 1981). The calculated statistics showed that the systematic component of the MSE was 99 and 93 % for the WE and Miglietta models, respectively, whereas for the modified WE model it was only 5 %.

DISCUSSION

Most wheat simulation models assume that only temperature and photoperiod affect LAR (e.g. Amir and Sinclair, 1991; Hodges and Ritchie, 1991; Wang and Engel, 1998). If this assumption is correct, then HS should increase linearly as a function of time (calendar days) under constant temperature and photoperiod. The results from the growth

chamber experiment (Fig. 3), where temperature and photoperiod were constant, showed that this assumption is not correct. Time plays an important role in modifying LAR throughout the growing season because seed reserves affect LAR for the first two leaves and later appearing leaves take more time to emerge from the whorl. These two factors were considered in the LAR chronology response function [eqn (5)], which described the change in LAR with time in both growth chamber and field experiments (Fig. 2). This LAR chronology response function was incorporated into an existing model (Wang and Engel, 1998), improving the predictions of HS under contrasting controlled-environment and field conditions (Figs 3 and 6). The low overall RMSE of 0.3 leaves obtained using the modified WE model in the field experiments (Fig. 6) represented a reduction of 57 and 73 % of the RMSE obtained with the WE model and with the Miglietta model, respectively.

The values for RMSE obtained using the WE model [eqn (1); 1.1 leaves for the growth chamber experiment and 0.7 leaves for the field experiments) are similar to those reported by Xue (2000) for the same cultivars grown at the same field location but in different growing seasons (1996–1997, 1997–1998, 1998–1999). Other models have also predicted the HS in wheat with a RMSE of about one leaf (Bindi *et al.*, 1995; Jamieson *et al.*, 1995, 1998). At low ($HS < 8$) and high ($HS > 12$) values of HS, the error with the WE model in our study was often greater than one leaf.

If the prediction of leaf number is out by one leaf in wheat, this difference can have a considerable impact on the predictions of other processes based on the number of emerged leaves. For example, assuming that a maximum of six green leaves can be present on a culm during the vegetative phase under non-limiting conditions of water and nitrogen (McMaster *et al.*, 1991), an error of one leaf represents an error of about 17 % in the prediction of leaf area (assuming all leaves have equal area). The error in leaf area is even larger when plants have fewer than six green leaves, a situation that is encountered at the beginning of the growing season or when plants undergo any biotic (e.g. pest or disease) or abiotic (e.g. water or N stress) stress (Wilhelm *et al.*, 1993). In the Sirius-Wheat model (Jamieson *et al.*, 1998), anthesis date occurs three phyllochrons after the emergence of the flag leaf. An error of one leaf in the prediction of the appearance of the flag leaf introduces an error in the prediction of anthesis that may correspond to several calendar days depending on temperature. In wheat, tillers appear at a predictable HS on the main stem (Klepper *et al.*, 1982, 1983) in an exponential fashion (Klepper *et al.*, 1982; Boone *et al.*, 1990; Rickman and Klepper, 1991). At the beginning of the growing season ($HS < 4$), the number of tillers is small and errors in the prediction of HS have a small impact on tiller appearance. However, at higher HS ($HS > 4$), an error of one leaf causes important errors in the prediction of tiller appearance. For instance, a plant with four, five and six fully expanded leaves can have three, four and eight tillers, respectively (Boone *et al.*, 1990; Rickman and Klepper, 1991). Errors in the predictions of tiller appearance will have further impacts on the predictions of appearance of leaves and growth of

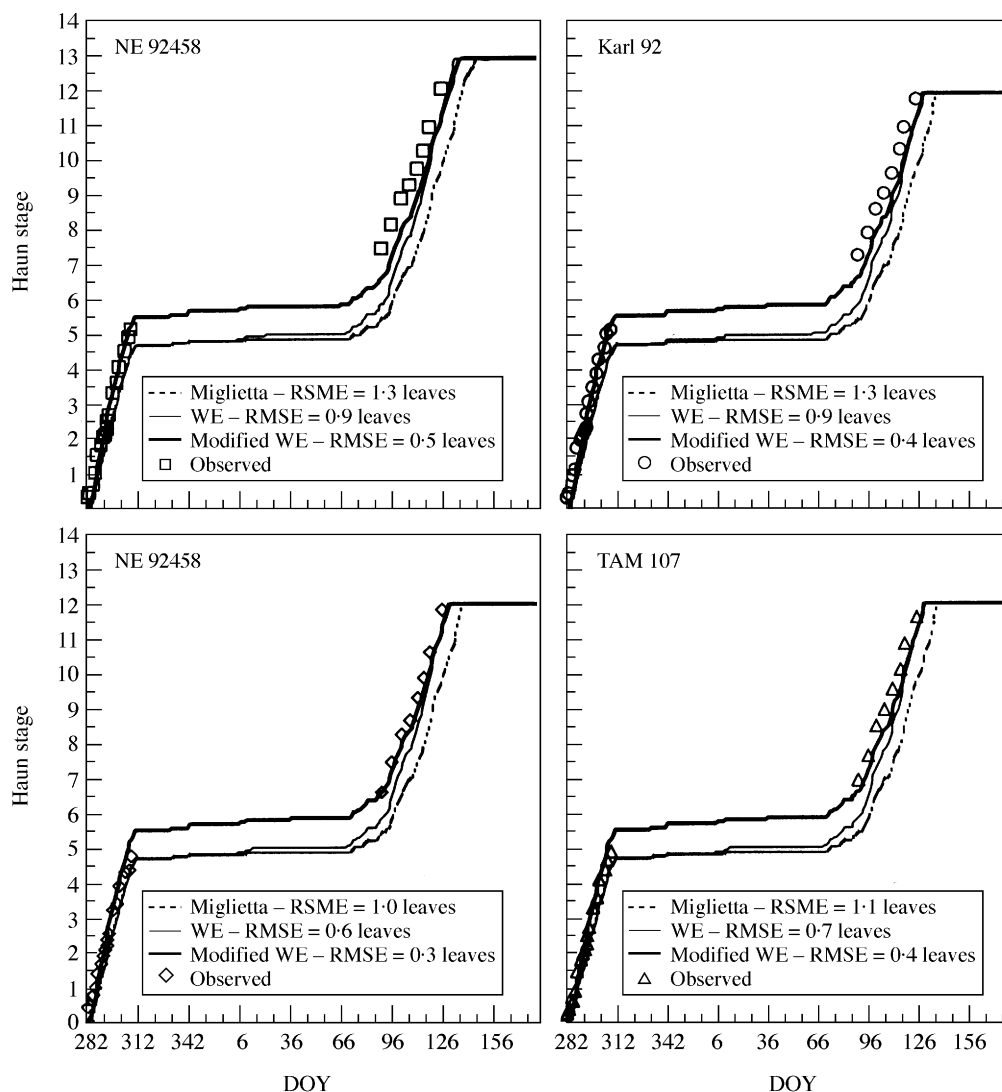


FIG. 5. Observed main stem Haun stage and values predicted by the three LAR models (Miglietta; WE; modified WE) for four winter wheat cultivars ('Arapahoe', 'Karl 92', 'NE92458' and 'TAM 107') sown on 29 Sep. 2000 (DOY = 273) in the field at Lincoln, NE, USA.

leaf area on tillers and, consequently, the whole plant leaf area.

Important features of any crop simulation model are generality and robustness whilst maintaining accurate predictions. The Miglietta model, which assumes a chronology effect on LAR, overpredicted HS in the growth chamber experiment (Fig. 3), and underpredicted HS most of the time under field conditions (Fig. 6). These differences in the predictive ability between controlled-environment and field conditions indicate a lack of generality in this model. There are several likely reasons for this lack of generality. First, the linear temperature function in the Miglietta model may not be entirely appropriate for all conditions. The response of LAR to temperature is linear only within a portion of the temperature range that affects LAR (Shaykewich, 1995; Jame *et al.*, 1999). Close to the minimum (or base) and optimum temperatures, the LAR response to temperature is non-linear (Cao and Moss, 1989a; Slafer and Rawson, 1995a; Hotsonyame and Hunt,

1997). Secondly, the Miglietta model has no photoperiod function. Photoperiod is also a major factor that affects LAR in wheat (Cao and Moss, 1989b; Slafer and Rawson, 1997). Thirdly, the Miglietta model is conceptually correct, but the chronology response function follows an exponential law and does not include seed reserves. Our results suggest that a power law function better describes the decrease in LAR with leaf number after seed reserves (Fig. 3). Fourthly, the coefficients of the Miglietta model may not be the most appropriate for the cultivars used in this study. What makes the coefficients of the Miglietta model difficult to determine for cultivars other than those used in the original study is that primordia number is required [eqn (7)]. Primordia number is much more difficult to measure than HS, and is therefore not usually available; we did not measure primordia number in this study.

The modified WE model performed excellently when predicting HS under a wide range of meteorological conditions for different sowing dates, different genotypes

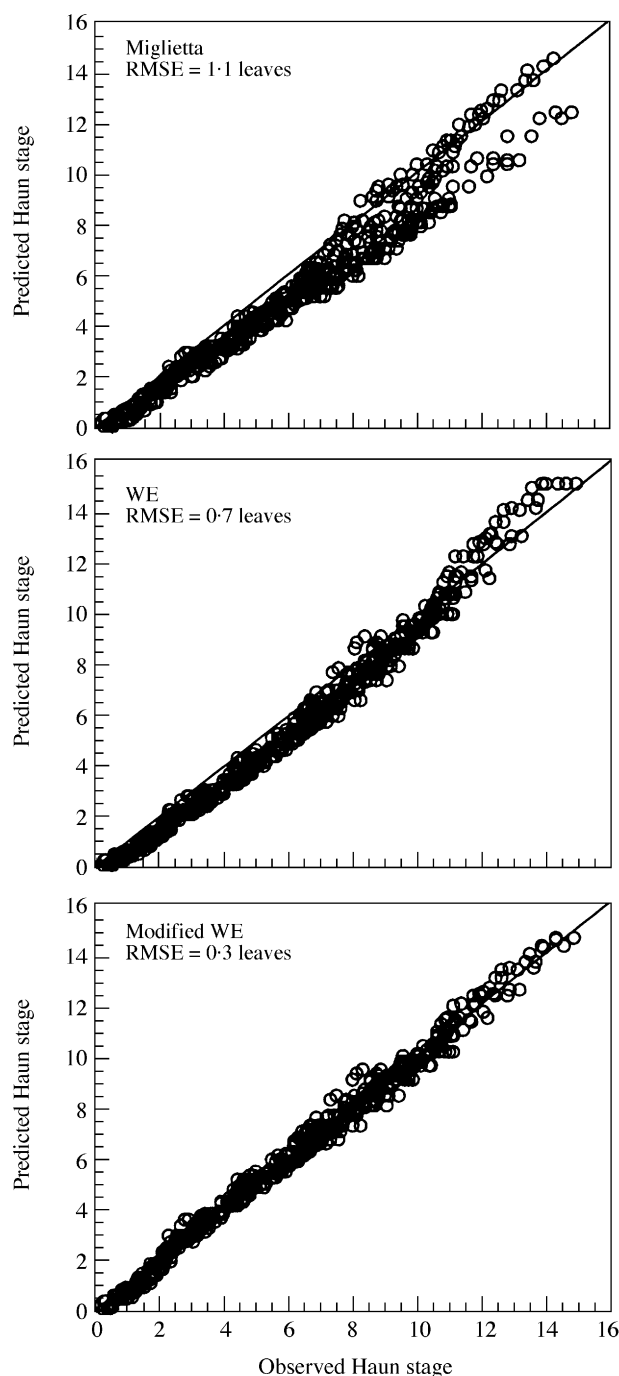


FIG. 6. Predicted vs. observed values for main stem Haun stage using the three LAR models (Miglietta; WE; modified WE). Data are pooled for four winter wheat cultivars ('Arapahoe', 'Karl 92', 'NE92458' and 'TAM 107') sown on eight dates in the field at Lincoln, NE, USA, during the 1999–2000 and 2000–2001 growing seasons. The solid line is the 1 : 1 line.

and different vernalization conditions. The modified WE model has a non-linear temperature function, a non-linear photoperiod function, and an LAR chronology function, with coefficients that have biological meaning. The value of $R_{LA,max}$ in the modified WE model [eqn (6)] was estimated

based on the value of $R_{LA,max}$ derived by Xue (2000), indicating that this procedure was reasonable and can be extended to other studies where $R_{LA,max}$ was estimated without seed reserves. The LAR chronology function [eqn (5)] introduced in the original WE model worked well for four winter wheat cultivars, indicating that this function can be considered independent of cultivar, and therefore no additional input is necessary in the modified WE model. The sensitivity coefficient (b) of the LAR chronology function was derived from another species (*Oryza sativa* L.) and worked well for wheat (Fig 2A and B). These results indicate the general and robust nature of the modified WE model, which constitutes an improvement over existing models.

CONCLUSIONS

Predictions of HS for different winter wheat cultivars using the WE model were greatly improved when a generalized LAR chronology response function [eqn (5)] was incorporated into the original model. The LAR chronology response function is realistic and biologically meaningful. It takes into account the effect of seed reserves on the rate of appearance of the first two leaves and the fact that higher leaves take more time to appear because the distance that each leaf primordium has to traverse to appear at the whorl increases for each subsequent leaf.

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